HOANG HUU BACH (M.Sc.)

Computer Science | **UET-VNU** | Hanoi, Vietnam | (+84)917 637 168
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Summary

Computer Science M.Sc. working across data engineering and applied machine learning. Build data pipelines, analytical models, dashboards, and sequence-learning systems for financial and scientific applications, combining industry data operations with research in cancer genomics, quantitative analytics, and university teaching.

Technical Skills

Data engineering: Python, pandas, SQL, DuckDB, Parquet, dbt; layered modeling, CTEs, joins, windows, ingestion, reconciliation, and quality tests.

Applied ML & retrieval: PyTorch, BLOSUM62, BiLSTM, cross-attention, Mixture-of-Experts, gating, ROC-AUC, ablations; fastText, word2vec.

Quantitative research: technical signals, simulated trading, walk-forward backtesting, QUBO, QAOA, Qiskit statevector simulation, COBYLA.

Analytics & visualization: Plotly, Dash, Tableau; feature engineering, exploratory analysis, statistics, and researcher-facing dashboards.

Software & data systems: Git, Bash, C/C++, Node.js, AngularJS, MongoDB/Mongoose, Elasticsearch; custom Python agents.

Experience

Aug 2023 – Present
◇ **Research Assistant & Teaching Assistant**, *UET-VNU*

+ Research bioinformatics and computational biology; develop and evaluate deep-learning methods for cancer genomics and peptide–HLA/neoantigen prediction.

+ Support *INT1008 — Introduction to Programming*, *INT2210 — Data Structures and Algorithms*, *COM1050 — Computational Thinking*, and *MAT1052/MAT1101 — Probability and Statistics*.

Feb 2024 – Aug 2024
◇ **Data Crawling Collaborator**, *LEXENGINE*

+ Crawled Vietnamese legal documents for LEXCentra, an AI-assisted retrieval system; analyzed document properties and used word2vec-based classification.

Oct 2023 – Feb 2024
◇ **Data Operations Intern**, *Singapore-based hedge fund*

+ Extracted and pre-processed data with Python/pandas and SQL according to researchers' requests; developed visualizations with Plotly and Tableau.

+ Deployed a Dash web dashboard presenting the visualizations and research outputs to researchers.

Jun 2023 – Nov 2023
◇ **Technical Support Specialist**, *Synodus JSC.*

+ Provided technical support to platform users, troubleshooting reported problems and escalating technical issues to developers; evaluated platform strengths and limitations; verified identities in mistaken-transaction cases to help prevent scams.

May 2022 – Sep 2022
◇ **AI Engineer Intern**, *GHTK Tech, NLP Department*

+ Applied fastText for semantic search and address retrieval; worked on address-search functionality and optimized suggested-search functions.

Education

2024 — 2026 **M.Sc.**, *Computer Science*, University of Engineering and Technology, Vietnam National University

Thesis: *Research on Deep Learning Models for neo-antigen detection and discovery* — 8.3/10.

Coursework: Statistical Machine Learning; Bioinformatics for Big Data; Algorithm Design and Analysis.

2019 — 2023 **B.Sc.**, *Computer Science, Honors Program*, UET-VNU

Thesis: *Calling somatic mutations from genomics next-generation sequencing data* — 9.7/10 (A+); *2023 Excellence Thesis Defense Certificate*.

Languages: Vietnamese — native; English — IELTS Academic 7.0 (23 July 2026).

Selected Data & AI Projects

Jun 2024 – Present

◇ **MHPE — Neoantigen Prediction with Mixture-of-Experts**

+ Designed and implemented peptide–HLA binding/presentation models for computational prioritization of neoantigen candidates.

+ Encoded sequences with BLOSUM62; developed BiLSTM representations and peptide–HLA cross-attention to model sequence context and input interactions.

+ Integrated multiple expert groups through a gating network to combine different biological signals in the prediction framework.

+ Compared accuracy and ROC-AUC; continuing ablations examine individual signal and expert contributions. Core framework reported at ICCBB 2025 (ACM). *Ongoing research.*

Jan 2025 – Present

◇ **BQuant — Quantitative Analytics, Agentic AI & Portfolio Optimization**

+ Build daily/15-minute Python ingestion into DuckDB and per-symbol Parquet, with stable-base/intraday-delta reconciliation, OHLC/volume/key checks, freshness manifests, and resilience tests.

+ Develop 16 dbt models (5 staging, 6 intermediate, 5 marts): CTEs, joins, lag/rolling windows for VN30 trend, momentum, volatility, liquidity, and benchmark-relative signals.

+ Implement simulated trading and walk-forward backtesting: prior-date signals, next-open execution, rebalancing, transaction costs, cash/board-lot checks, and settlement-aware orders.

+ Integrate custom Python agents with warehouse context, persisted knowledge, recommendation records, and reporting, using technical-analysis and QAOA evidence.

+ Formulate binary asset selection as QUBO with heuristic-alpha, covariance-risk, turnover, and position-count terms; prototype QAOA with Qiskit statevector/COBYLA and exact classical fallback, then derive capped allocation weights with a cash buffer.

`github.com/h2bach/BQuant` — *ongoing personal project.*

May 2025 – Jun 2026

◇ **BiodiversityVN — Phylogenetic Data Platform**

+ Extended an existing Node.js/AngularJS platform for database adaptation and phylogenetic inference: MongoDB species/sequence records, CSV/accession mapping, record merging, and Elasticsearch retrieval.

+ Integrated FASTA/BLAST search, MUSCLE alignment, and IQ-TREE output. Ministry of Science and Technology project CN.3780. `github.com/h2bach/dnatracker_clone`.

Publications

[C1] **Bach Huu Hoang** and Thanh Hai Dang. *Neo-antigen prediction using Mixture of Experts for HLA-peptides interactions with multi-factors incorporated. Proceedings of the 2025 9th International Conference on Computational Biology and Bioinformatics (ICCBB 2025)*, pp. 76–81. ACM, 2025. DOI: 10.1145/3789938.3789948.

[C2] **Hoang Huu Bach**, Duc Kien Nguyen, and Nghiem Nguyen Viet Dung. *Quantum Approach for Constructing Phylogenetic Maximum Parsimony Tree. Future Data and Security Engineering (FDSE 2024)*, CCIS vol. 2310, pp. 158–170. Springer, 2024. DOI: 10.1007/978-981-96-0437-1_12.

[C3] **Hoang Huu Bach**, Nguyen Thi Thuong, and Son Thanh Cao. *An Empirical Heuristic Algorithm for Solving the Student-Project Allocation Problem with Ties. Proceedings of the 15th International Conference on Agents and Artificial Intelligence (ICAART 2023)*, vol. 3, pp. 665–672, 2023. DOI: 10.5220/0011731800003393.

[C4] Nguyen Van Duc, **Hoang Huu Bach**, and Le Hong Trang. *An Improved Reverse Distillation Model for Unsupervised Anomaly Detection. Proceedings of the 17th International Conference on Ubiquitous Information Management and Communication (IMCOM 2023)*, pp. 1–6. IEEE, 2023. DOI: 10.1109/IMCOM56909.2023.10035610.